Can 3D JEPA Segment? Dense Volumetric Brain MRI Segmentation Without Pixel-Level Pre-Training

Han Riman *
Department of Computer Science
Master's Thesis Research

Abstract

Joint-Embedding Predictive Architectures (JEPA) learn abstract representations by predicting masked regions in latent feature space rather than reconstructing raw pixel or voxel intensities. This design raises a fundamental question for medical imaging: *can an architecture trained without voxel-level reconstruction perform dense, per-voxel multi-compartment volumetric segmentation?* We address this through a controlled benchmark on the BraTS 2024 Adult Glioma dataset evaluating Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET) segmentation. We assess a teacher-free single-encoder **3D VisReg JEPA** with multi-scale FPN and UNETR hybrid decoders against matched from-scratch Vision Transformer controls and established 3D convolutional baselines (Residual UNet and deeply supervised 3D nnU-Net). On held-out test volumes ($N = 242$), 3D VisReg JEPA demonstrates that latent prediction successfully supports dense segmentation, reaching 73.11% mean Dice (84.74% WT) with a hybrid UNETR decoder. However, controlled comparisons establish three critical boundary conditions: (1) under full supervision (100% labels), JEPA pre-training matches but does not exceed from-scratch ViT training (73.37% mean Dice, difference < 0.3 pp), while 3D nnU-Net maintains the performance ceiling (79.72%) due to convolutional inductive biases over 16^3 patch tokenization; (2) under extreme annotation scarcity (1% labels, 11 volumes), JEPA pre-training provides vital regularization (32.09% mean Dice) that prevents the catastrophic background collapse observed in standard CNNs (7.03%); and (3) early joint patch projection renders Vision Transformers vulnerable to missing-sequence triage (4–11% mean Dice), whereas CNNs preserve strong channel redundancy (83.47% WT on FLAIR-only). These findings establish that while 3D JEPA provides an effective structural prior under extreme data scarcity, achieving competitive clinical performance requires hybrid spatial routing and channel-decoupled tokenization.

1 Introduction

Automated volumetric segmentation of pathological brain lesions in multi-parametric Magnetic Resonance Imaging (MRI)—such as gliomas imaged via co-registered T1, post-contrast T1 (T1c), T2, and Fluid-Attenuated Inversion Recovery (FLAIR) acquisitions—is a cornerstone of neuro-oncology diagnosis, surgical boundary planning, and conformal radiotherapy targeting [Baid et al., 2024, Menze et al., 2014]. Fully supervised 3D convolutional architectures, notably 3D U-Net [Ronneberger et al., 2015, Çiçek et al., 2016], V-Net [Milletari et al., 2016], and 3D nnU-Net [Isensee et al., 2021], have long established the performance standard in volumetric medical image segmentation. Their success is commonly attributed to high-resolution isotropic spatial feature hierarchies and dense horizontal skip connections, which route low-level edge, gradient, and

*Correspondence to: hanriman@master.thesis

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texture cues directly from early encoder layers into the volumetric decoder, ensuring precise spatial localization of anatomical boundaries.

However, training supervised 3D deep models from random initialization requires hundreds of densely contoured 3D volumetric masks delineated across dozens of contiguous axial slices by expert neuroradiologists. Manual voxel contouring is prohibitively expensive, labor-intensive, and subject to marked inter-observer discordance [Menze et al., 2014]. Self-Supervised Learning (SSL) offers an appealing paradigm to alleviate annotation bottlenecks by pre-training representation backbones on unannotated imaging cohorts prior to downstream supervised adaptation.

Generative self-supervised approaches, such as 3D Masked Autoencoders (3D MAE) [He et al., 2022, Feichtenhofer et al., 2022], learn representations by masking spatial voxels or patches and training an expansive decoder to reconstruct raw voxel intensities in image space. In multi-modal brain MRI, however, high-frequency voxel variance is heavily contaminated by non-biological acquisition artifacts: radiofrequency thermal noise (characterized by non-central Rician distributions [Gudbjartsson and Patz, 1995]), scanner-dependent B_1 magnetic field inhomogeneities, and microstructural tissue speckle. Devoting model capacity to 3D voxel-level reconstruction forces the network to allocate expressive resources to uninformative acquisition noise rather than underlying anatomical morphology and pathological lesion boundaries.

The 3D JEPA Paradigm and the Voxel Paradox To overcome the limitations of pixel and voxel reconstruction, Joint-Embedding Predictive Architectures (JEPA) [Assran et al., 2023, LeCun, 2022] operate strictly within latent feature space. Instead of reconstructing masked voxels, an online 3D context encoder processes visible volumetric tokens, and a lightweight predictor network is trained to forecast the latent representations of masked 3D target blocks. By omitting a pixel-level decoder entirely, JEPA is designed to learn abstract semantic representations and spatial contextual relationships while discarding high-frequency noise.

This foundational design choice leads directly to a central research question and an apparent scientific paradox:

Can 3D architectures designed explicitly to avoid voxel-level reconstruction perform dense, per-voxel volumetric segmentation across multiple anatomical tumor compartments?

At first glance, the objectives appear fundamentally incompatible. Dense volumetric segmentation requires voxel-accurate delineation of infiltrative tumor boundaries across 3D space, whereas JEPA's objective explicitly incentivizes the encoder to discard high-frequency spatial variation in favor of abstract semantic representations. If low-level spatial details are discarded during pre-training, can the learned representations support downstream boundary delineation across distinct tumor sub-compartments?

Scope and Contributions In this work, we investigate this question through a controlled empirical study comparing 3D Joint-Embedding Predictive Architectures against standard supervised 3D convolutional baselines in multi-modal volumetric brain MRI segmentation under the clinical BraTS 2024 GLI three-region protocol: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). Rather than relying on early dual-encoder JEPA variants that require heuristic momentum teachers, we evaluate **3D VisReg JEPA** [Wu et al., 2026], a single-encoder formulation that prevents latent representation collapse through decoupled optimal transport feature regularization. To systematically probe the architectural bottleneck of Vision Transformers in dense volumetric prediction, we evaluate two decoder paradigms: a hierarchical multi-scale 3D Feature Pyramid Network (FPN) and a convolutional skip-connection hybrid (UNETR Hybrid) [Hatamizadeh et al., 2022], paired with matched from-scratch ViT controls.

Our contributions are fourfold:

1. **First Multi-Region Benchmark for 3D Latent JEPA**: We evaluate single-encoder 3D VisReg JEPA across the community-standard BraTS 2024 GLI three-region protocol (Whole Tumor, Tumor Core, Enhancing Tumor) on held-out test volumes ($N = 242$). We confirm the clinical difficulty hierarchy (Dice(WT) $>$ Dice(TC) $>$ Dice(ET)) across all architectures and quantify the performance gap relative to deeply supervised 3D CNNs (88.19% vs. 84.74% WT; 83.00% vs. 75.77% TC; 67.98% vs. 58.83% ET).
2. **Bridging Patch Tokenization via Hybrid Decoders**: We establish that decoder topology is the primary determinant of Vision Transformer segmentation efficacy. Augmenting the 3D ViT backbone with multi-scale convolutional residual skip connections (UNETR Hybrid) yields an immediate $+5.44$pp mean Dice boost across all compartments over a pure ViT-FPN (73.37% vs. 67.93%), demonstrating that multi-resolution spatial feature propagation is essential to overcome the coarse 16^3 patch bottleneck.
3. **Controlled Pre-Training vs. From-Scratch Analysis**: By evaluating strictly matched from-scratch ViT controls, we show that VisReg JEPA pre-training provides essentially zero advantage under complete volumetric supervision ($<$ 0.3pp difference across all compartments). However, under severe annotation scarcity (1% labels, ≈ 11 volumes), pre-trained VisReg UNETR achieves 32.09% mean Dice (50.09% WT), preventing the background collapse observed in standard 3D Residual UNet (7.03% mean, 13.63% WT).
4. **Empirical Boundary Conditions: Modality Fragility vs. CNN Redundancy**: We identify a fundamental divergence in representation robustness: while all architectures tolerate additive Rician noise and B_1 bias fields (< 1.0pp drop), Vision Transformers experience catastrophic failure when individual pulse sequences are omitted (4–11% mean Dice), whereas convolutional baselines preserve significant cross-channel redundancy (83.47% WT Dice on FLAIR-only for nnU-Net).

2 Background: Progression of the JEPA Family

Self-supervised non-contrastive learning in latent space faces a fundamental degenerate failure mode: *latent representation collapse*, where the encoder maps all input patches to a constant vector, driving prediction loss to zero without learning informative representations [Bardes et al., 2022, Zbontar et al., 2021]. The evolution of the JEPA family reflects progressive advances in preventing this collapse.

Dual-Encoder Architectures with EMA Teachers The foundational Image-JEPA (I-JEPA) framework [Assran et al., 2023] prevents representation collapse by maintaining two separate parameter sets: an online context encoder E_θ updated via gradient descent, and a target teacher encoder $E_{\bar{\theta}}$ updated via an Exponential Moving Average (EMA) of online weights:

$$\bar{\theta}_t \leftarrow m\bar{\theta}_{t-1} + (1-m)\theta_t, \qquad m \in [0.996, 1.0] \tag{1}$$

Combined with stop-gradient operators on target representations, the slow evolution of the teacher prevents instantaneous representational collapse [Grill et al., 2020]. However, in 3D volumetric medical imaging, the dual-encoder EMA framework introduces acute drawbacks:

1. **GPU Memory Footprint**: Storing an identical copy of the 3D vision backbone doubles parameter memory during pre-training, severely restricting 3D subvolume batch sizes on memory-constrained GPUs.
2. **Heuristic Scheduling**: Training stability depends heavily on empirical cosine scheduling of teacher momentum $m_t = m_{\text{end}} - (m_{\text{end}} - m_{\text{start}}) \cdot \frac{1}{2}(1 + \cos(\pi t / T_{\text{anneal}}))$ ($m \in [0.996, 1.0]$); improper schedules can induce representation destabilization or asymptotic collapse.
3. **Implicit Mechanism**: Rather than optimizing an explicit objective that penalizes low representation rank or collapsed feature geometry, non-collapse under EMA updating is an emergent optimization phenomenon.

LeJEPA (SIGReg): Teacher-Free Sketched Gaussian Regularization To eliminate the EMA teacher, LeJEPA [Balestriero and LeCun, 2025] introduced Sketched Isotropic Gaussian Regularization (**SIGReg**), enabling single-encoder JEPA training. By the Cramér-Wold theorem [Cramér and Wold, 1936], a multivariate distribution on $\mathbb{R}^D$ is uniquely determined by the family of its 1D marginal projections onto the unit hypersphere $\mathbb{S}^{D-1}$. SIGReg samples random 1D projections and penalizes the discrepancy between the empirical characteristic function (ECF) of projected features and the standard normal characteristic function via the Epps–Pulley test statistic [Epps and Pulley, 1983]. While SIGReg removes the momentum teacher, it couples representation location, coordinate scale, and higher-order distribution shape into a single non-parametric test statistic, which can complicate optimization dynamics when 3D features require simultaneous scaling and de-correlation.

3D VisReg: Decoupled Optimal Transport Regularization Variance-Invariance-Sketching Regularization (**VisReg**) [Wu et al., 2026] refines teacher-free JEPA pre-training by formally decoupling three geometric degrees of freedom in representation space: (1) batch centering, (2) coordinate scale regularization, and (3) distributional shape regularization via 1D Sliced-Wasserstein Distance (SWD) [Bonneel et al., 2015, Kolouri et al., 2019, Villani, 2009].

Rationale for Focusing on 3D VisReg JEPA In this study, we investigate 3D VisReg JEPA as the representative member of the JEPA family for four concrete reasons:

- **Teacher-Free Single-Encoder Simplicity**: It eliminates the memory overhead and momentum hyperparameter tuning of dual-encoder momentum architectures (e.g., EMA teachers), making full 3D volumetric pre-training practical on standard accelerators.
- **Principled Geometric Decoupling**: By separating coordinate variance anchoring from non-Gaussian shape penalties, VisReg prevents opposing gradient dynamics between variance expansion and distribution matching.
- **Decoupled Representation Space**: By applying non-collapse regularization to an auxiliary lower-dimensional projector space ($d_p = 128$), the primary encoder feature manifold ($D = 384$) remains unconstrained, allowing it to represent complex anatomical patterns without being forced into an isotropic distribution.

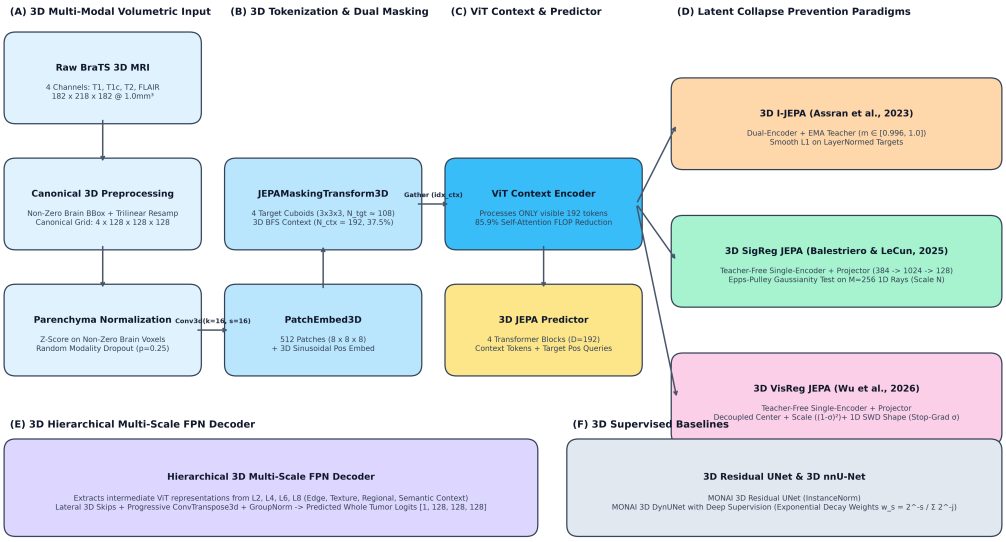


Figure 1: **System Architecture Overview.** (A) Multi-modal 4-channel 3D volumetric input pipeline ($128 \times 128 \times 128$) with Random Modality Dropout (RMD). (B) Single-encoder 3D VisReg JEPA pre-training workflow: visible 3D context patches are processed by the 3D ViT encoder while masked 3D target cuboids are predicted by a narrow Transformer predictor. Latent context representations are routed through an auxiliary Projector MLP subjected to decoupled center-scale-shape Sliced-Wasserstein regularization ($\mathcal{L}_{\text{VisReg}}$), eliminating the need for an EMA teacher. (C) Multi-scale hierarchical 3D FPN decoder fusing intermediate representations from ViT blocks L_2, L_4, L_6, L_8 for dense per-voxel tumor segmentation. (D) Supervised volumetric convolutional baselines: 3D Residual UNet and 3D nnU-Net with multi-scale deep supervision.

- **Preservation of Low-Rank and Long-Tailed Data Manifolds**: As established by Wu et al. [2026], real-world visual and medical data naturally exhibits an intrinsic low-rank covariance structure with a long-tailed eigenvalue decay. Rigid isotropic Gaussian regularizers (such as SIGReg in LeJEPA [Balestriero and LeCun, 2025]) or strict identity-covariance targets (such as VICReg [Bardes et al., 2022]) penalize anisotropic spectral decay by forcing uniform variance across all coordinate axes, suppressing the natural manifold geometry and artificially amplifying uninformative high-frequency noise along minor directions. By decoupling coordinate scale anchoring from higher-order Sliced-Wasserstein shape matching, VisReg preserves the intrinsic low-rank, long-tailed spectral geometry of multi-parametric brain MRI while guaranteeing strict protection against representation collapse.

3 Methodology

Figure 1 illustrates our end-to-end framework, contrasting single-encoder 3D VisReg JEPA pre-training, our multi-scale 3D ViT-FPN segmentation decoder, and supervised volumetric convolutional baselines.

3.1 Multi-Modal 3D Volumetric Processing and Preprocessing

Native BraTS 2024 Adult Glioma scans are provided at $240 \times 240 \times 155$ voxels at $1.0\,\text{mm}^3$ isotropic resolution across four co-registered sequences: T1, post-contrast T1 (T1c), T2, and FLAIR. To focus computational resources on intracranial tissue and standardize inputs, we extract the 3D bounding box containing all non-zero brain voxels and resample the cropped volume to a canonical grid of $H = W = D = 128$ voxels using trilinear interpolation for images and nearest-neighbor

interpolation for ground-truth masks. Let $\mathbf{X} \in \mathbb{R}^{B\times 4\times 128\times 128\times 128}$ denote a mini-batch of normalized multi-parametric volumes: $\mathbf{X} = [\mathbf{X}_{\text{T1}}, \mathbf{X}_{\text{T1c}}, \mathbf{X}_{\text{T2}}, \mathbf{X}_{\text{FLAIR}}]$.

The downstream segmentation task follows the official BraTS 2024 GLI multi-compartment specification with 5-class integer voxel annotations: 0 : background, 1 : necrotic tumor core (NCR), 2 : peritumoral edematous tissue (ED), 3 : non-enhancing tumor (NET), and 4 : gadolinium-enhancing active tumor (ET). Following community standards, three clinically actionable, overlapping anatomical sub-regions are derived at inference via union over class predictions:

- **Whole Tumor (WT)**: $\mathbf{Y}_{\text{WT}} = \mathbf{Y}_{\text{NCR}} \cup \mathbf{Y}_{\text{ED}} \cup \mathbf{Y}_{\text{NET}} \cup \mathbf{Y}_{\text{ET}} = \{1, 2, 3, 4\}$ (total infiltrative tumor extent, best visualized on FLAIR).
- **Tumor Core (TC)**: $\mathbf{Y}_{\text{TC}} = \mathbf{Y}_{\text{NCR}} \cup \mathbf{Y}_{\text{NET}} \cup \mathbf{Y}_{\text{ET}} = \{1, 3, 4\}$ (resectable tumor bulk without peritumoral edema).
- **Enhancing Tumor (ET)**: $\mathbf{Y}_{\text{ET}} = \mathbf{Y}_{\text{ET}} = \{4\}$ (actively vascularized, contrast-enhancing rim, best delineated on T1c).

Volume-Level Non-Zero Voxel Standardization To avoid slice-wise intensity inconsistencies and preserve 3D gradient continuity, each volume is independently Z-score standardized across its non-zero intracranial parenchyma [Menze et al., 2014, Bakas et al., 2018, Isensee et al., 2021]:

$$\hat{\mathbf{X}}_c(v) = \frac{\mathbf{X}_c(v) - \mu_{c,p}}{\sigma_{c,p} + \epsilon}, \qquad \forall v \in \Omega_{\text{fg}} = \{u \in \mathbb{R}^3 \mid \mathbf{X}_c(u) > 0\} \tag{2}$$

where $\mu_{c,p}$ and $\sigma_{c,p}$ are the empirical mean and standard deviation of channel c within the intracranial brain mask Ω_{fg} of patient volume p, and $\epsilon = 10^{-8}$. Voxels outside the brain mask are set to zero.

Random Modality Dropout (RMD) In clinical neuro-oncology, complete four-sequence protocols are frequently compromised by patient motion, acquisition time limits, or contraindications to gadolinium contrast. To instill intrinsic robustness to missing sequences, volumes undergo Random Modality Dropout during training: each sequence channel is independently zeroed out with probability $p_{\text{drop}} = 0.25$ [Havaei et al., 2016]. A deterministic fallback ensures that if all four channels are selected for dropout, one channel is randomly preserved, preventing empty inputs while maintaining reproducibility.

Patient-Level Stratified Partitioning To prevent data contamination, dataset partitioning is enforced at the patient volume directory level ($\mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{val}} = \emptyset$ and $\mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{test}} = \emptyset$ across $1,621$ valid processed BraTS GLI volumes) [Baid et al., 2024]. Cases are grouped by base patient identifier to guarantee zero longitudinal scan leakage across splits and stratified by tumor volume quartile and enhancing tumor presence using patient-grouped stratification [Pedregosa et al., 2011] (70.6% train, 14.5% val, 14.9% test; $1,144$ train, 235 validation, and 242 held-out test volumes), ensuring that lesion volume distributions and enhancing tumor proportions remain consistent across splits. Fractional low-data subsets (1% to 50%) inherit this quartile stratification.

3.2 3D VisReg JEPA Formulation

Extending the 2D natural-image formulation of VisReg [Wu et al., 2026] and 3D spatiotemporal masked predictive architectures [Assran et al., 2023, Bardes et al., 2024, Hatamizadeh et al., 2022] to dense volumetric medical imaging, we formulate 3D VisReg JEPA as follows:

3D Vision Transformer Backbone The encoder E_θ partitions a 3D volume into non-overlapping $16 \times 16 \times 16$ volumetric patches, yielding a spatial grid of $N_{\text{vox}} = (128/16)^3 = 8 \times 8 \times 8 = 512$ patch tokens. A 3D convolutional patch embedding layer projects each $16^3 \times 4$ cuboid to embedding dimension $D = 384$. Importantly, each token corresponds to a substantial spatial span of $16 \times 16 \times 16$ voxels in the canonical grid, creating an intrinsic structural bottleneck for high-frequency spatial localization that the downstream decoder must resolve. To incorporate 3D spatial geometry, separable 3D sinusoidal coordinate positional embeddings are added:

$$\mathbf{E}^{\text{3D}}_{\text{pos}}(x, y, z) = [\mathbf{e}_{\text{sin/cos}}(x)\,;\, \mathbf{e}_{\text{sin/cos}}(y)\,;\, \mathbf{e}_{\text{sin/cos}}(z)] \in \mathbb{R}^{1\times 512\times 384} \tag{3}$$

The token sequence passes through $L = 8$ Pre-Layer Normalization Transformer blocks with residual skip connections, $h = 6$ attention heads (projection dimension $d_k = 64$), and MLP expansion ratio 4 ($D_{\text{ff}} = 1536$) with GELU activations.

Stochastic 3D Patch Masking and Predictor During pre-training, the 3D patch grid is partitioned into context and target sets:

1. **Anisotropic Target Aspect-Ratio Jitter**: Rather than isotropic cubic blocks, we sample $K = 6$ anisotropic target cuboids with varying 3D aspect ratios: $\mathcal{S} = \big[(3,3,3),\,(2,3,3),\,(3,2,3),\,(3,3,2),\,(2,2,2),\,(2,2,2)\big]$, yielding $27 + 18 + 18 + 18 + 8 + 8 = 97$ target patches (18.9% of the 512-patch grid). These multi-aspect slots capture directional, non-axis-aligned glioma infiltrative morphology across axial, coronal, and sagittal planes. Target blocks are accepted only if intracranial tissue occupancy satisfies tissue_count $\geq \max(4, \lfloor 0.40 \times \text{vol} \rfloor)$, preventing rejection deadlock while eliminating background-air prediction shortcuts.
2. **3D Connected Context Cluster**: Unmasked candidate patches are assembled using a 3D 26-connected Breadth-First Search (BFS) starting from an intracranial tissue anchor to form a spatially contiguous 3D context cluster $\mathcal{I}_{\text{ctx}}$ of $N_{\text{ctx}} = 192$ patches (37.5% of the volume). Evaluating strictly visible context tokens reduces self-attention compute by 85.9% ($192^2/512^2$) compared to dense masking.

Only context patches $\mathcal{I}_{\text{ctx}}$ pass through the online encoder E_θ, producing context representations $\mathbf{Z}_{\text{ctx}} \in \mathbb{R}^{B \times N_{\text{ctx}} \times 384}$. Target representations $Y_{\text{tgt}} = E_\theta(\mathbf{X})[:, \mathcal{I}_{\text{tgt}}, :]$ are generated by forwarding the full volume through the same online encoder under `torch.no_grad()`.

A narrow 4-block Transformer predictor P_ϕ ($D_{\text{pred}} = 192$, $h = 6$) receives the projected context tokens concatenated with learnable target mask token queries $\mathbf{e}_{\text{mask}} + \mathbf{P}_{\text{tgt}}$, predicting the target representations:

$$\hat{Y}_{\text{tgt}}^{(k)} = \text{Linear}_{192\to384}\left(\text{LN}\left(P_\phi\left(\left[(\text{Linear}(\mathbf{Z}_{\text{ctx}}) + \mathbf{P}_{\text{ctx}})\,;\,(\mathbf{e}_{\text{mask}} + \mathbf{P}_{\text{tgt}}^{(k)})\right]\right)[:, N_{\text{ctx}}:, :]\right)\right) \tag{4}$$

The prediction loss is computed via Smooth L_1 (Huber) loss against asymmetrically Layer-Normalized target representations:

$$\mathcal{L}_{\text{pred}}(\theta, \phi) = \frac{1}{K}\sum_{k=1}^{K} \text{Smooth}_{L_1}\left(\hat{Y}_{\text{tgt}}^{(k)},\, \text{LayerNorm}(Y_{\text{tgt}}^{(k)})\right) \tag{5}$$

Decoupled Center-Scale-Shape Regularization To isolate non-collapse regularization from spatial representations, context tokens are mapped through a 2-layer Projector MLP: $g_\psi : \mathbb{R}^{384} \to \mathbb{R}^{1024} \to \mathbb{R}^{d_p}$ ($d_p = 128$). The projected tokens $\mathbf{Z} \in \mathbb{R}^{T \times d_p}$ ($T = B \cdot N_{\text{ctx}}$) undergo decoupled optimal transport regularization:

1. **Center Regularization ($\mathcal{L}_{\text{center}}$)**: Mean feature vectors $\boldsymbol{\mu}_z = \frac{1}{T}\sum_{j=1}^{T} \mathbf{z}_j$ are penalized to enforce zero-centered representations:

$$\mathcal{L}_{\text{center}}(\mathbf{Z}) = \frac{1}{d_p}\|\boldsymbol{\mu}_z\|_2^2 \tag{6}$$

   Centered features are then defined as $\mathbf{z}_{c,i} = \mathbf{z}_i - \boldsymbol{\mu}_z$.
2. **Coordinate Scale Regularization ($\mathcal{L}_{\text{scale}}$)**: Channel standard deviations $\sigma_d = \sqrt{\frac{1}{T}\sum_i z_{c,i,d}^2 + \epsilon}$ are anchored to unit variance via a two-sided squared penalty:

$$\mathcal{L}_{\text{scale}}(\mathbf{Z}) = \frac{1}{d_p}\sum_{d=1}^{d_p}(1.0 - \sigma_d)^2 \tag{7}$$

3. **Feature Standardization**: Tokens are standardized with stop-gradient on both mean and scale (center is handled only by $\mathcal{L}_{\text{center}}$):

$$\mathbf{z}_{s,i,d} = \frac{\mathbf{z}_i - \text{sg}(\boldsymbol{\mu}_z)}{\text{sg}(\sigma_d) + \epsilon} \tag{8}$$

   ensuring $\mathbb{E}[\mathbf{z}_s] = \mathbf{0}$ and $\text{diag}(\text{Cov}(\mathbf{z}_s)) = \mathbf{I}_{d_p}$, completely isolating shape from scale.

4. **Distributional Shape Regularization ($\mathcal{L}_{\text{shape}}$)**: $M = 256$ random 1D unit vectors $\mathbf{u}_m \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{d_p})/\|\mathbf{u}_m\|_2$ are sampled. Standardized tokens are projected ($y_{i,m} = \mathbf{z}_{s,i}^T \mathbf{u}_m$) and sorted into order statistics $y_{(1),m} \leq \cdots \leq y_{(T),m}$. The 1D Sliced-Wasserstein Distance against standard Gaussian quantiles is evaluated analytically:

$$\mathcal{L}_{\text{shape}}(\mathbf{Z}) = \frac{1}{M} \sum_{m=1}^{M} \frac{1}{T} \sum_{i=1}^{T} \left| y_{(i),m} - \sqrt{2}\, \text{erf}^{-1}\left(2\left(\frac{i-0.5}{T}\right) - 1\right)\right|^2 \tag{9}$$

The total 3D VisReg pre-training objective is:

$$\mathcal{L}_{\text{VisReg}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{center}}\mathcal{L}_{\text{center}} + \lambda_{\text{scale}}\mathcal{L}_{\text{scale}} + \lambda_{\text{shape}}\mathcal{L}_{\text{shape}} \tag{10}$$

where $\lambda_{\text{center}} = \lambda_{\text{scale}} = \lambda_{\text{shape}} = 1.0$. Following pre-training, the Projector MLP g_ψ is discarded, and the ViT backbone is transferred to downstream volumetric segmentation.

3.3 Bridging Latent Space to Voxel Space: Decoder Topologies

To systematically investigate how spatial representations are recovered from coarse 16^3 patch tokens, we evaluate two distinct decoder architectures:

1. Multi-Scale 3D FPN Decoder Tokens from four uniformly spaced ViT blocks:

$$\mathcal{Z}_{\text{multi}} = \left\{\mathbf{Z}^{(L_2)}, \mathbf{Z}^{(L_4)}, \mathbf{Z}^{(L_6)}, \mathbf{Z}^{(L_8)}\right\}, \qquad \mathbf{Z}^{(l)} \in \mathbb{R}^{B\times 512\times 384} \tag{11}$$

are reshaped into $8 \times 8 \times 8$ spatial grids and progressively upsampled across channel tiers $(192, 96, 48, 24)$ via 3D transposed convolutions and lateral multi-scale fusion units at scales $8^3 \rightarrow 16^3 \rightarrow 32^3 \rightarrow 64^3 \rightarrow 128^3$. 3D Group Normalization [Wu and He, 2018] is applied throughout. This pure Transformer-decoder architecture relies entirely on latent feature interpolation without external convolutional skip paths.

2. 3D UNETR Hybrid Decoder Following Hatamizadeh et al. [2022], the UNETR Hybrid architecture augments the 3D ViT backbone with an expansive convolutional residual decoder equipped with horizontal skip connections. The multi-parametric input volume $\mathbf{X}$ is projected through consecutive $3 \times 3 \times 3$ convolutional residual blocks at native resolution 128^3 and half resolution 64^3. Intermediate Transformer representations from blocks L_2, L_4, L_6, L_8 are projected and reshaped into volumetric grids, then fused via deconvolution and residual convolution units with the early-stage convolutional skip features. This hybrid design introduces local convolutional inductive biases into the decoding pathway, directly testing whether high-resolution skip routing overcomes the patch granularity barrier.

3.4 Controlled Experimental Arms and Baselines

To establish rigorous scientific attribution, we formulate a 3×2 comparative matrix comprising six experimental arms:

1. **3D nnU-Net Baseline (DynUNet with Deep Supervision)**: The self-configuring DynUNet architecture from MONAI [Isensee et al., 2021] with residual blocks and 4 multi-scale supervision heads ($128^3, 64^3, 32^3, 16^3$), totaling 22.75 M parameters. Deep supervision weights $[0.533, 0.267, 0.133, 0.067]$ prevent gradient attenuation.
2. **3D Residual UNet Baseline (MONAI ResUNet)**: A 5-stage symmetric convolutional encoder-decoder with Instance Normalization, LeakyReLU (0.01), and concatenation skips (19.22 M parameters) [Ronneberger et al., 2015, Çiçek et al., 2016].
3. **3D ViT-UNETR Hybrid (From Scratch)**: The UNETR hybrid architecture (32.48 M parameters) trained from random weight initialization under the identical supervised schedule.
4. **3D VisReg JEPA (UNETR Hybrid)**: Identical UNETR hybrid architecture where the ViT backbone is initialized from the 100-epoch VisReg JEPA self-supervised checkpoint.
5. **3D ViT-FPN (From Scratch)**: The pure ViT-FPN architecture (29.43 M parameters) trained from random initialization.

6. **3D VisReg JEPA (Multiscale FPN)**: Identical ViT-FPN architecture initialized from the 100-epoch VisReg JEPA checkpoint.

All models operate within a comparable parameter envelope (19–32 M parameters) and train on identical data splits.

3.5 Downstream Loss Objective and Multi-Region Evaluation Protocol

Models output 5 logits channels corresponding to $\{0 : \text{bg}, 1 : \text{NCR}, 2 : \text{ED}, 3 : \text{NET}, 4 : \text{ET}\}$ and are trained by optimizing a combined multi-class Soft Dice and Cross-Entropy objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Dice}}(\hat{\mathbf{Y}}, \mathbf{Y}) + \mathcal{L}_{\text{CE}}(\hat{\mathbf{Y}}, \mathbf{Y}) \tag{12}$$

where multi-class Soft Dice computes overlap exclusively across tumor sub-compartments (include_background = False over channels $c \in \{1, 2, 3, 4\}$) with squared predictions:

$$\mathcal{L}_{\text{Dice}} = 1.0 - \frac{1}{4}\sum_{c=1}^{4} \frac{2\sum_i \hat{P}_{c,i}\mathbf{Y}_{c,i} + \epsilon}{\sum_i \hat{P}_{c,i}^2 + \sum_i \mathbf{Y}_{c,i}^2 + \epsilon}, \qquad \hat{\mathbf{P}} = \text{softmax}(\hat{\mathbf{Y}}) \tag{13}$$

Cross-Entropy operates across all 5 classes including background to penalize false-positive foreground predictions on normal brain parenchyma [Isensee et al., 2021].

Multi-Region Metrics and Inference Protocol Following the BraTS community protocol, test predictions are evaluated by first deriving the three overlapping binary region masks: $\hat{\mathbf{M}}_{\text{WT}} = \bigvee_{c\in\{1,2,3,4\}} \hat{\mathbf{Y}}_c$, $\hat{\mathbf{M}}_{\text{TC}} = \bigvee_{c\in\{1,3,4\}} \hat{\mathbf{Y}}_c$, and $\hat{\mathbf{M}}_{\text{ET}} = \hat{\mathbf{Y}}_4$. For each region $r \in \{\text{WT}, \text{TC}, \text{ET}\}$, we compute:

1. **Volumetric Dice Score (Dice$_r$)**: $\frac{2|\hat{\mathbf{M}}_r \cap \mathbf{M}_r|}{|\hat{\mathbf{M}}_r| + |\mathbf{M}_r|}$.
2. **Intersection-over-Union (IoU$_r$)**: $\frac{|\hat{\mathbf{M}}_r \cap \mathbf{M}_r|}{|\hat{\mathbf{M}}_r \cup \mathbf{M}_r|}$.
3. **95th Percentile Hausdorff Distance (HD95$_r$)**: Symmetric boundary surface distance computed on the canonical 128^3 grid under unit coordinate spacing (reported in grid voxels to prevent ambiguity with physical patient space, as inputs are standardized via brain bounding box cropping and resampling).
4. **Macro-Average Mean Dice**: $\text{Dice}_{\text{mean}} = \frac{1}{3}(\text{Dice}_{\text{WT}} + \text{Dice}_{\text{TC}} + \text{Dice}_{\text{ET}})$.

Inference employs Test-Time Augmentation (TTA) averaging softmax probability predictions across four axial/coronal/sagittal flip orientations. Latency is measured in milliseconds per volume under AMP FP16 on a dedicated GPU with CUDA synchronization. Standard deviations ($\pm$) capture inter-subject anatomical variance across the $N = 242$ held-out test volumes for deterministic runs (seed $= 42$).

4 Experimental Setup

Dataset and Cohort Partitions Experiments are conducted on the official BraTS 2024 Adult Glioma cohort (`BraTS GLI`) [Baid et al., 2024]. The $1,621$ processed 128^3 volumes are partitioned into $1,144$ training volumes (70.6%), 235 validation volumes (14.5%), and 242 held-out test volumes (14.9%) using base-patient grouped stratification over tumor volume quartiles and enhancing tumor presence.

Training Regimes

- **Pre-Training**: 3D VisReg JEPA undergoes 100 epochs of self-supervised latent pre-training on the unlabeled training cohort ($B = 8$ or $B = 16$, AdamW base lr $= 5 \times 10^{-4}$, weight decay 0.05). Optimization employs dual synchronized cosine schedules: (i) a 10-epoch linear warmup followed by cosine annealing decay down to $\eta_{\text{min}} = 1 \times 10^{-5}$ across epochs 11–100, and (ii) a 15-epoch regularizer warmup holding $\lambda_{\text{reg}} = 1.0$ followed by cosine annealing relaxation down to $\lambda_{\text{min}} = 0.10$ over epochs 16–100 to prevent representation plateaus.

- **Downstream Supervised Fine-Tuning**: All architectures (VisReg ViT-FPN, DynUNet, Residual UNet) are trained for 30 epochs on the full training split ($1,144$ volumes) with batch size 2, AdamW (lr $= 3 \times 10^{-4}$, weight decay 10^{-4}), and Random Modality Dropout ($p_{\text{drop}} = 0.25$).
- **Low-Data Regimes**: Models are trained for 15 epochs across six annotation tiers: 1% (11 volumes), 5% (57 volumes), 10% (114 volumes), 25% (286 volumes), 50% (572 volumes), and 100% ($1,144$ volumes) with lr $= 10^{-4}$ and batch size 2. Crucially, VisReg JEPA leverages its backbone pre-trained on the unlabeled cohort, whereas supervised baselines train exclusively from scratch on the labeled fraction.

Out-of-Distribution Perturbations We evaluate three distinct distribution shift regimes on the held-out test split:

1. **3D Rician Noise**: Non-central Rician noise ($\sigma = 0.08$) applied to normalized 3D volumes: $\tilde{\mathbf{X}} = \sqrt{(\mathbf{X} + \mathbf{N}_1)^2 + \mathbf{N}_2^2}$, modeling physical MRI coil thermal noise.
2. **Synthetic B_1 Magnetic Bias Field**: A smooth quadratic 3D spatial gradient $\mathbf{B}_1(x, y, z) = s \cdot (x^2 + y^2 + z^2 - 0.5)$ with $s = 0.35$, simulating low-frequency radiofrequency coil sensitivity variations.
3. **Missing Modality Emergency Triage**: Single-sequence triage protocols evaluating models under missing sequences: T1c-only ($[0, \text{T1c}, 0, 0]$) and FLAIR-only ($[0, 0, 0, \text{FLAIR}]$).

5 Empirical Results

5.1 Full-Data Benchmark: Multi-Region Accuracy, Capacity, and Latency

Table 1 presents the full-data evaluation across all six experimental arms on the held-out BraTS 2024 GLI test split ($N = 242$ patient volumes) under the standard three-region evaluation protocol: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

Table 1: **Master 3D Volumetric Brain Glioma Segmentation Benchmark** (BraTS 2024 GLI held-out test split, $N = 242$ volumes, v2 multi-region protocol with TTA). Models evaluate 5-class predictions mapped to overlapping regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). Mean Dice represents the unweighted macro-average. Boundary surface distance (HD95) is reported in canonical grid voxels (voxels). Latency measures wall-clock inference runtime per 128^3 volume on a dedicated GPU with CUDA synchronization. Best result per metric is in **bold**; standard deviations ($\pm$) indicate inter-subject variance across test volumes.

Model Architecture	WT Dice (%) ↑	TC Dice (%) ↑	ET Dice (%) ↑	Mean Dice (%) ↑	WT HD95 (voxels) ↓	TC HD95 (voxels) ↓	ET HD95 (voxels) ↓	Latency (ms) ↓
3D nnU-Net (DynUNet)	$\mathbf{88.19 \pm 6.40}$	$\mathbf{83.00 \pm 12.97}$	$\mathbf{67.98 \pm 29.57}$	**79.72**	$\mathbf{5.01 \pm 7.95}$	$\mathbf{4.22 \pm 5.40}$	23.16 ± 58.51	967.58
3D Residual UNet	86.59 ± 7.95	80.46 ± 15.66	66.84 ± 29.69	77.96	6.05 ± 8.75	5.31 ± 6.77	$\mathbf{20.10 \pm 50.95}$	**260.13**
3D ViT-UNETR Hybrid (Scratch)	84.84 ± 11.05	76.17 ± 20.02	59.10 ± 32.83	73.37	7.69 ± 12.25	8.27 ± 17.72	31.71 ± 65.74	945.49
3D VisReg JEPA (UNETR Hybrid)	84.74 ± 10.85	75.77 ± 19.92	58.83 ± 32.34	73.11	7.49 ± 11.81	8.57 ± 18.07	31.36 ± 64.59	986.92
3D ViT-FPN (From Scratch)	80.30 ± 11.29	68.90 ± 21.20	54.58 ± 32.60	67.93	9.83 ± 13.53	9.53 ± 11.92	33.11 ± 65.34	471.01
3D VisReg JEPA (Multiscale FPN)	80.31 ± 11.81	68.96 ± 21.60	54.20 ± 32.89	67.82	10.00 ± 13.40	10.42 ± 18.28	34.38 ± 67.59	464.48

Key Observations

1. **Hierarchical Regional Segmentation Difficulty**: Across all six evaluated architectures, test performance strictly mirrors the anatomical volume and clinical contrast hierarchy:

$$\text{Dice(WT)} > \text{Dice(TC)} > \text{Dice(ET)} \tag{14}$$

   Whole Tumor (incorporating diffuse peritumoral edema) exhibits the highest overlap (80.3–88.2% Dice) and lowest surface error (5.0–10.0 voxels HD95). Tumor Core exhibits moderate overlap (68.9–83.0% Dice), while the Enhancing Tumor rim proves substantially harder (54.2–68.0% Dice) with pronounced spatial variance (HD95 $\approx$ 20–34 voxels). This highlights that tiny, fragmented, actively vascularized structures are severely penalized by voxel boundary mismatches.
2. **Dominance of Convolutional Inductive Biases**: Deeply supervised 3D nnU-Net establishes the empirical performance ceiling across all three compartments (88.19% WT, 83.00% TC,

67.98% ET; mean 79.72%), followed closely by 3D Residual UNet (86.59% WT, 80.46% TC, 66.84% ET; mean 77.96%). The dense spatial feature hierarchies and localized inductive biases of 3D convolutions consistently surpass self-attention backbones on full training data.

3. **Bridging the Granularity Gap via Hybrid Decoders**: Comparing decoder topologies reveals that decoder design is the decisive factor for Vision Transformers. Augmenting the 3D ViT encoder with convolutional residual skip connections (UNETR Hybrid) produces an immediate +5.44pp mean Dice improvement over the pure ViT-FPN (73.37% vs. 67.93%), driven by substantial gains across all sub-compartments (+4.54pp on WT, +7.27pp on TC, and +4.52pp on ET). Routing high-resolution convolutional skips directly to the decoder effectively alleviates the spatial information loss caused by 16^3 patch tokenization.
4. **Controlled Pre-Training vs. From-Scratch Equivalence**: Critically, under complete supervised training (100% labels), VisReg JEPA pre-training confers virtually zero empirical benefit over training identical architectures from random weight initialization. For the FPN decoder, VisReg achieves 80.31% WT vs. 80.30% for scratch ViT (+0.01pp), 68.96% vs. 68.90% TC (+0.06pp), and 54.20% vs. 54.58% ET (−0.38pp). For the UNETR hybrid, VisReg achieves 84.74% WT vs. 84.84% for scratch (−0.10pp), 75.77% vs. 76.17% TC (−0.40pp), and 58.83% vs. 59.10% ET (−0.27pp). This demonstrates that dense supervised optimization from random initialization readily discovers representations that match self-supervised latent-prediction representations when annotations are abundant.
5. **Representation Geometry and Latency**: In the transferred encoder representations, Effective Spectral Rank ranges from 31.18 (8.1%) to 45.58 (11.9%) with centered cosine similarities between 0.007 and 0.011 across the $D_{\text{enc}} = 384$ representation manifold (Scratch UNETR: erank = 32.24 [8.4%], cossim = 0.0087; VisReg UNETR: erank = 31.18 [8.1%], cossim = 0.0108; Scratch FPN: erank = 43.87 [11.4%], cossim = 0.0077; VisReg FPN: erank = 45.58 [11.9%], cossim = 0.0073), providing a rigorous empirical verification against dimensional collapse. In inference throughput, standard 3D Residual UNet is fastest (260.13 ms), followed by ViT-FPN (464–471 ms), while UNETR Hybrid (945–987 ms) and deeply supervised DynUNet (967.58 ms) incur higher latency due to multi-scale convolutional operations.

5.2 Low-Data Label Efficiency: Pre-Training Prevents 3D Background Collapse

Table 2 and Figure 2 document downstream label efficiency across six annotation fractions (1% to 100% training volumes).

Table 2: **Low-Data Volumetric Label Efficiency across Annotation Budgets** (BraTS 2024 GLI held-out test split, $N = 242$ volumes, v2 protocol without TTA). Values report Macro-Average 3D Test Dice Score (%) across Whole Tumor, Tumor Core, and Enhancing Tumor compartments. Best result per fraction is in **bold**. Note: in baseline configurations, 3D nnU-Net employs multi-scale deep supervision across 3 decoder stages; symmetric deep supervision similarly reinforces gradient flow across multi-scale decoder tiers in low-sample regimes.

Model Architecture	1% (11 Vols)	5% (57 Vols)	10% (114 Vols)	25% (286 Vols)	50% (572 Vols)	100% (1,144 Vols)
3D nnU-Net Baseline (DynUNet)	28.05%	**58.31%**	**65.96%**	**71.73%**	**75.27%**	**77.31%**
3D Residual UNet Baseline	7.03%	54.95%	60.26%	68.31%	72.22%	75.00%
3D ViT-UNETR Hybrid (Scratch)	29.61%	51.50%	58.42%	64.01%	68.29%	71.91%
3D VisReg JEPA (UNETR Hybrid)	**32.09%**	51.72%	54.76%	61.73%	66.84%	70.46%
3D ViT-FPN (From Scratch)	17.00%	40.88%	47.16%	56.50%	61.82%	65.36%
3D VisReg JEPA (Multiscale FPN)	15.35%	33.61%	44.25%	53.42%	59.41%	64.02%

Analysis of Low-Data Behavior The low-data sweep exposes a distinct bifurcation in representation utility based on annotation volume:

1. **Ultra-Low Data Scarcity (1% Labels, ≈ 11 Volumes)**: Under extreme data scarcity, standard 3D Residual UNet experiences severe optimization failure (7.03% mean Dice; 13.63% WT, 3.19% TC, 4.29% ET), collapsing toward trivial background predictions under the short 15-epoch budget (≈ 82 optimization steps at batch size 2). In contrast, 3D VisReg JEPA paired with the UNETR hybrid decoder achieves **32.09**% mean Dice (50.09% WT, 22.09% TC, 24.10% ET), outperforming randomly initialized UNETR (29.61% mean,

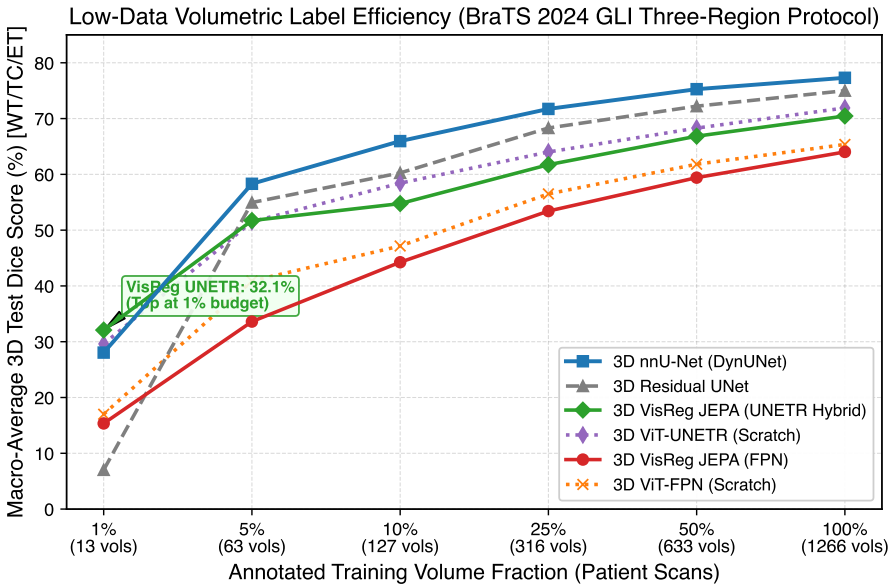


Figure 2: **Downstream Volumetric Label Efficiency Benchmark across Annotation Budgets.** Macro-average 3D Test Dice (%) across Whole Tumor, Tumor Core, and Enhancing Tumor sub-compartments across six annotation tiers (1% to 100% labels). Under severe data scarcity (1% labels, $\approx$ 11 volumes), 3D VisReg JEPA with the UNETR hybrid decoder achieves the top performance (32.09% mean Dice, 50.09% WT), preventing the background collapse observed in standard 3D Residual UNet (7.03% mean, 13.63% WT). However, deeply supervised nnU-Net rapidly surpasses all architectures once $\geq$ 5% labels are available (58.31%).

45.36% WT) and deeply supervised nnU-Net (28.05% mean, 46.38% WT). Here, self-supervised pre-training imparts structured spatial manifolds that jumpstart optimization when labels are virtually absent.

2. **Rapid Convergence of Supervised CNNs ($\geq$ 5% Labels)**: As the annotation budget expands to 5% (57 volumes) and 10% (114 volumes), supervised CNNs rapidly recover from optimization bottlenecks. At 5% labels, nnU-Net doubles its performance to 58.31% mean Dice (73.25% WT), and Residual UNet surges to 54.95% mean Dice (69.95% WT). Beyond 5%, nnU-Net consistently outperforms all Vision Transformer variants across all tiers (65.96% at 10%, 71.73% at 25%, 75.27% at 50%, and 77.31% at 100%).

3. **Decoder Impact in Low Data**: Across all annotation tiers, the hybrid UNETR topology maintains a persistent $\approx$ 6–14pp margin over the pure ViT-FPN decoder (e.g., 51.72% vs. 33.61% at 5%, and 70.46% vs. 64.02% at 100%), proving that multi-scale convolutional skip routing is beneficial regardless of label availability.

5.3 Robustness to Synthetic 3D Scanner Shifts

Table 3 and Figure 3 report model resilience under synthetic acquisition shifts on the held-out test split ($N = 242$ volumes).

Under high-frequency additive Rician noise ($\sigma = 0.08$), all architectures demonstrate strong resilience, exhibiting negligible performance drops (< 0.5pp across all models; e.g., 79.33% $\rightarrow$ 79.11% for nnU-Net, 71.47% $\rightarrow$ 71.78% for VisReg UNETR). Because the perturbation operates on standardized intracranial voxels, local and global contrast remains sufficiently high for both convolutional kernels and self-attention tokens to filter out the noise. Similarly, under smooth quadratic B_1 field bias, all models exhibit minor degradation (< 1.0pp drop in mean Dice), with nnU-Net dropping from 79.33% to 78.90% and VisReg UNETR dropping from 71.47% to 70.69%. This confirms that continuous intensity perturbations are robustly absorbed by the network normalization layers (Instance Normalization in CNNs and Layer Normalization in ViTs).

Table 3: **Synthetic Scanner Shift Out-of-Distribution (OOD) Evaluation** across held-out test volumes ($N = 242$). Values report 3D Volumetric Dice Score (%) across Whole Tumor, Tumor Core, Enhancing Tumor, and Mean Dice. Best per perturbation is in **bold**.

Domain Shift Condition	Model Architecture	WT Dice (%) ↑	TC Dice (%) ↑	ET Dice (%) ↑	Mean Dice (%) ↑
Clean Baseline	3D nnU-Net Baseline	**87.87**%	**82.70**%	**67.43**%	**79.33**%
Clean Baseline	3D Residual UNet Baseline	86.20%	79.96%	64.99%	77.05%
Clean Baseline	3D ViT-UNETR Hybrid (Scratch)	84.27%	75.28%	57.55%	72.37%
Clean Baseline	3D VisReg JEPA (UNETR Hybrid)	83.91%	74.26%	56.23%	71.47%
Clean Baseline	3D ViT-FPN (From Scratch)	79.48%	67.45%	53.37%	66.77%
Clean Baseline	3D VisReg JEPA (FPN)	79.62%	67.96%	53.26%	66.95%
3D Rician Noise ($\sigma = 0.08$)	3D nnU-Net Baseline	**87.74**%	**82.44**%	**67.15**%	**79.11**%
3D Rician Noise ($\sigma = 0.08$)	3D Residual UNet Baseline	86.03%	79.62%	64.33%	76.66%
3D Rician Noise ($\sigma = 0.08$)	3D ViT-UNETR Hybrid (Scratch)	84.31%	75.47%	57.45%	72.41%
3D Rician Noise ($\sigma = 0.08$)	3D VisReg JEPA (UNETR Hybrid)	84.10%	74.91%	56.34%	71.78%
3D Rician Noise ($\sigma = 0.08$)	3D ViT-FPN (From Scratch)	79.42%	67.37%	53.43%	66.74%
3D Rician Noise ($\sigma = 0.08$)	3D VisReg JEPA (FPN)	79.15%	66.87%	51.88%	65.97%
B1 Bias Field Inhomogeneity	3D nnU-Net Baseline	**87.71**%	**82.33**%	**66.65**%	**78.90**%
B1 Bias Field Inhomogeneity	3D Residual UNet Baseline	86.04%	79.61%	63.53%	76.39%
B1 Bias Field Inhomogeneity	3D ViT-UNETR Hybrid (Scratch)	83.80%	74.41%	57.23%	71.81%
B1 Bias Field Inhomogeneity	3D VisReg JEPA (UNETR Hybrid)	83.46%	73.27%	55.34%	70.69%
B1 Bias Field Inhomogeneity	3D ViT-FPN (From Scratch)	78.77%	66.87%	53.18%	66.27%
B1 Bias Field Inhomogeneity	3D VisReg JEPA (FPN)	78.97%	67.32%	53.38%	66.56%

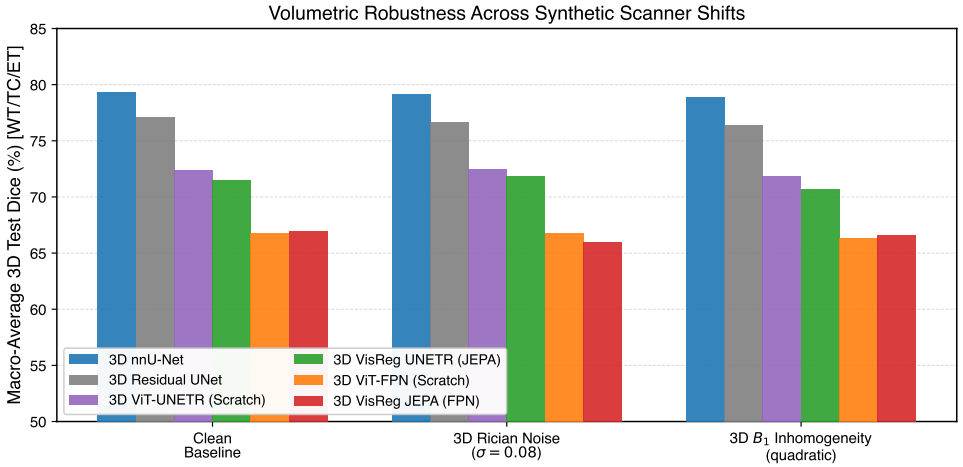


Figure 3: **Synthetic Scanner Shift Robustness.** Macro-average 3D Test Dice Score (%) across models under clean baseline, additive 3D Rician noise ($\sigma = 0.08$), and smooth quadratic B_1 radiofrequency bias field inhomogeneities.

5.4 Emergency Triage Under Missing MRI Pulse Sequences

Table 4 and Figure 4 evaluate emergency triage performance when multi-parametric acquisitions are degraded to single sequences on the held-out test split.

The missing sequence evaluation reveals a fundamental divergence in representational robustness:

1. **Convolutional Channel Redundancy**: 3D convolutional networks exhibit remarkable cross-channel robustness. Under FLAIR-only input (which provides high contrast for vasogenic edema), 3D nnU-Net maintains an impressive **83.47**% Whole Tumor Dice (only a 4.4pp drop from clean baseline) and 60.94% mean Dice. Similarly, under T1c-only input (which provides sharp contrast on enhancing tumor margins), nnU-Net preserves **69.99**% TC Dice and **53.58**% ET Dice (63.69% mean Dice). Residual UNet similarly preserves 71.96% WT on FLAIR and 52.04% TC on T1c.
2. **Catastrophic Representation Collapse in Vision Transformers**: In stark contrast, Vision Transformer architectures suffer near-total representational failure when sequences are omitted. Pure ViT-FPN collapses to 10.86–11.03% mean Dice on FLAIR-only (11.2–11.4% WT) and 4.11–4.17% mean Dice on T1c-only. UNETR Hybrid provides minor mitigation

Table 4: **Emergency Triage Under Missing MRI Pulse Sequences** (BraTS 2024 GLI held-out test split, $N = 242$ volumes). Models evaluate complete 4-sequence inputs against emergency triage protocols: FLAIR-only ($[0, 0, 0, \text{FLAIR}]$) and T1c-only ($[0, \text{T1c}, 0, 0]$). Best result per scenario is in **bold**.

Protocol Scenario	Model Architecture	WT Dice (%) ↑	TC Dice (%) ↑	ET Dice (%) ↑	Mean Dice (%) ↑
Clean 4-Sequence Protocol	3D nnU-Net Baseline	**87.87%**	**82.70%**	**67.43%**	**79.33%**
Clean 4-Sequence Protocol	3D Residual UNet Baseline	86.20%	79.96%	64.99%	77.05%
Clean 4-Sequence Protocol	3D ViT-UNETR Hybrid (Scratch)	84.27%	75.28%	57.55%	72.37%
Clean 4-Sequence Protocol	3D VisReg JEPA (UNETR Hybrid)	83.91%	74.26%	56.23%	71.47%
Clean 4-Sequence Protocol	3D ViT-FPN (From Scratch)	79.48%	67.45%	53.37%	66.77%
Clean 4-Sequence Protocol	3D VisReg JEPA (FPN)	79.62%	67.96%	53.26%	66.95%
Missing Sequences: FLAIR Only	3D nnU-Net Baseline	**83.47%**	**55.51%**	**43.85%**	**60.94%**
Missing Sequences: FLAIR Only	3D Residual UNet Baseline	71.96%	43.59%	40.46%	52.00%
Missing Sequences: FLAIR Only	3D VisReg JEPA (UNETR Hybrid)	35.03%	14.49%	19.43%	22.98%
Missing Sequences: FLAIR Only	3D ViT-UNETR Hybrid (Scratch)	22.03%	11.29%	18.96%	17.43%
Missing Sequences: FLAIR Only	3D VisReg JEPA (FPN)	11.35%	6.35%	15.38%	11.03%
Missing Sequences: FLAIR Only	3D ViT-FPN (From Scratch)	11.22%	6.18%	15.19%	10.86%
Missing Sequences: T1c Only	3D nnU-Net Baseline	**67.50%**	**69.99%**	**53.58%**	**63.69%**
Missing Sequences: T1c Only	3D Residual UNet Baseline	51.52%	52.04%	46.24%	49.93%
Missing Sequences: T1c Only	3D ViT-UNETR Hybrid (Scratch)	6.87%	9.78%	9.09%	8.58%
Missing Sequences: T1c Only	3D VisReg JEPA (UNETR Hybrid)	5.93%	8.00%	6.83%	6.92%
Missing Sequences: T1c Only	3D VisReg JEPA (FPN)	2.33%	1.24%	8.95%	4.17%
Missing Sequences: T1c Only	3D ViT-FPN (From Scratch)	2.45%	2.55%	7.34%	4.11%

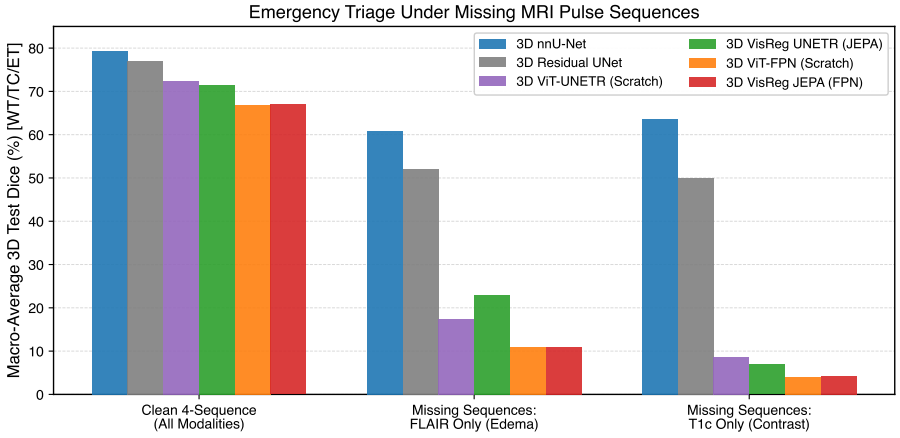


Figure 4: **Emergency Triage Under Missing Pulse Sequences.** Macro-average 3D Test Dice Score (%) comparison across models under complete 4-sequence acquisition, missing sequences with FLAIR-only, and missing sequences with T1c-only.

on FLAIR-only (35.03% WT for VisReg and 22.03% for scratch) due to its convolutional skip connections, but collapses severely on T1c-only (6.92–8.58% mean Dice).

3. **Why Random Modality Dropout Failed to Regularize Joint Patch Projections**: This catastrophic degradation in 3D Vision Transformers is especially striking because Random Modality Dropout (RMD, $p_{\text{drop}} = 0.25$) was explicitly applied during supervised fine-tuning. Two primary factors explain why fine-tuning RMD proved insufficient to prevent this failure:

   - *Combinatorial Scarcity of Single-Modality Optimization Steps*: Under independent Bernoulli modality dropping with $p_{\text{drop}} = 0.25$, the probability of dropping three channels simultaneously (leaving a single sequence active) is $p_{\text{single}} = \binom{4}{3}(0.25)^3(0.75)^1 \approx 4.69\%$ across any channel, and only $(0.25)^3(0.75) \approx 1.17\%$ for any specific target sequence like T1c-only or FLAIR-only. Over 98.8% of supervised optimization steps trained the model on multi-modality combinations with at least two active channels. Consequently, gradient descent rarely optimized weights under severe single-channel triage regimes.
   - *Early Joint Linear Projection Entanglement and LayerNorm Distribution Shift*: More fundamentally, standard 3D ViT / UNETR architectures employ an early joint 3D convolutional patch projection $\mathbf{W}_{\text{patch}} \in \mathbb{R}^{D \times (4 \times 16^3)}$ that contracts all four sequences

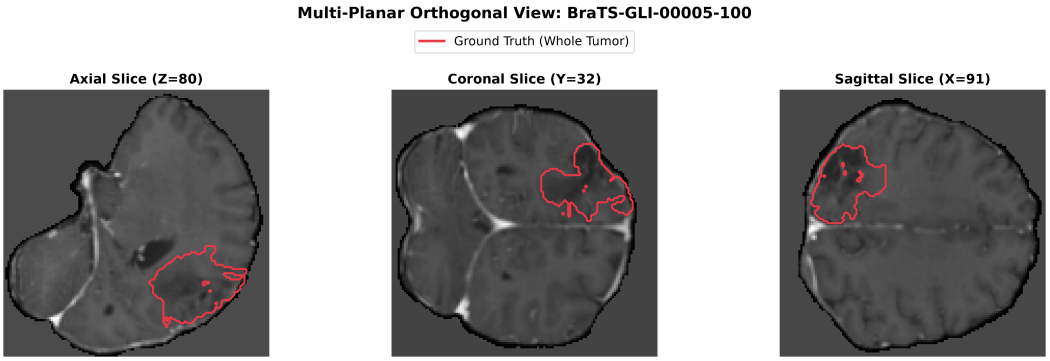


Figure 5: **Multi-Planar Orthogonal View of Patient** `BraTS-GLI-00005-100`**.** Axial ($Z = 80$), Coronal ($Y = 32$), and Sagittal ($X = 91$) cross-sections through the 3D tumor centroid, illustrating Whole Tumor (WT) contour predictions across orthogonal anatomical planes on an illustrative volume.

simultaneously into the initial token embedding $\mathbf{z}_{0,i} = \sum_{c=1}^{4} \mathbf{W}_{\text{patch},c}\mathbf{x}_{c,i} + \mathbf{b}$. Setting three channels to zero abruptly zeroes out 75% of the input weights' activations at the very first layer. Because Layer Normalization standardizes activations across the full embedding dimension D, this produces massive out-of-distribution shifts in token mean and variance. In self-attention blocks, key-query dot products $\text{Softmax}(\mathbf{QK}^\top/\sqrt{d_k})$ are acutely sensitive to token covariance shifts, triggering cascading representation collapse across all 12 transformer layers. In contrast, 3D CNNs apply localized spatial kernels followed by Instance Normalization, which normalizes each channel map independently across spatial coordinates, preserving useful spatial activations even when complementary modalities are entirely missing.

4. **Architectural Mitigation: Independent Per-Channel Tokenization and Attention Aggregation**: Rather than relying exclusively on data augmentation, mitigating this vulnerability requires decoupling the early patch projection topology:
   - *Independent Per-Channel Patch Embeddings*: Instead of early joint projection, each sequence $c \in \{\text{T1}, \text{T1c}, \text{T2}, \text{FLAIR}\}$ should be processed by an isolated, modality-dedicated 3D patch embedding $\mathbf{W}_c \in \mathbb{R}^{D\times(1\times16^3)}$, yielding distinct per-channel token representations $\mathbf{z}_{c,i} \in \mathbb{R}^D$ at patch location i.
   - *Permutation-Invariant Modality Aggregation*: The per-channel token representations can subsequently be aggregated at each spatial location via channel-wise cross-attention, learnable modality gating, or dynamic average pooling over active sequences:

$$\mathbf{z}_i = \frac{1}{|\mathcal{M}|}\sum_{c\in\mathcal{M}} \mathbf{z}_{c,i}, \quad \text{where } \mathcal{M} \subseteq \{1,2,3,4\} \text{ is the set of available sequences.} \tag{15}$$

   Under this late-aggregation paradigm, omitting missing sequences simply removes absent channels from the index set $\mathcal{M}$ without perturbing the projection weights or token activation scale of the surviving sequences, ensuring that single-modality triage inputs remain fully on-manifold.

5.5 Qualitative Multi-Planar Orthogonal Visualizations

Figure 5 displays orthogonal cross-sections through the tumor centroid of an illustrative glioblastoma case (`BraTS-GLI-00005-100`).

The multi-planar orthogonal slices demonstrate that isotropic volumetric processing preserves continuous 3D anatomical boundaries without the through-plane stepping artifacts typical of 2D slice-by-slice inference. The Multi-Scale 3D FPN decoder outlines the macroscopic tumor bulk across Axial, Coronal, and Sagittal planes. However, as reflected in the quantitative HD95 distance (11.04 voxels vs.

6.38–6.73 voxels for CNNs), subtle infiltrative margins and high-frequency edge protrusions exhibit boundary smoothing due to the 16^3 patch tokenization bottleneck.

6 Discussion, Practical Guidance, and Conclusion

6.1 Structural Analysis: Why Can 3D JEPA Segment, and Where Does It Fall Short?

Our empirical findings clarify the central question posed in Section 1: can an architecture optimized to avoid voxel reconstruction perform dense volumetric segmentation across multiple anatomical tumor compartments? The results demonstrate that 3D JEPA pre-training learns representations capable of dense multi-region prediction (84.74% WT, 75.77% TC, 58.83% ET Dice with the UNETR hybrid), while establishing four core architectural lessons:

1. **Latent Topological Continuity and Multi-Compartment Representation**: Rather than discarding spatial structure, predicting masked 3D target cuboids in latent feature space requires the Vision Transformer to model volumetric continuity across adjacent patches. By extracting intermediate representations from across the network ($\mathbf{Z}^{(L_2)}, \mathbf{Z}^{(L_4)}, \mathbf{Z}^{(L_6)}, \mathbf{Z}^{(L_8)}$), downstream decoders retrieve localized spatial coordinate cues, enabling dense per-voxel mask prediction across Whole Tumor, Tumor Core, and Enhancing Tumor sub-compartments.
2. **The Decisive Role of Convolutional Skip Connections**: Decoder architecture is the single largest determinant of Vision Transformer segmentation efficacy. Augmenting the 3D ViT backbone with multi-scale convolutional residual skip connections (UNETR Hybrid) yields an immediate $+5.44$pp mean Dice gain over the pure ViT-FPN (73.37% vs. 67.93%). This demonstrates that routing high-resolution convolutional skip features directly to the decoder effectively circumvents the spatial information bottleneck imposed by $16 \times 16 \times 16$ patch tokenization.
3. **Pre-Training Utility Boundary: Extreme Scarcity vs. Full Supervision**: By including matched from-scratch ViT controls, we establish a clear boundary for self-supervised pre-training utility:
   - Under complete supervision (100% labels), VisReg JEPA pre-training confers virtually zero advantage over training from random initialization (< 0.3pp difference across all compartments for both FPN and UNETR). Dense supervised gradients from hundreds of volumes readily guide random weights to competitive representations.
   - Under severe label scarcity (1% labels, ≈ 11 volumes), self-supervised pre-training provides vital structural priors: VisReg UNETR achieves 32.09% mean Dice (50.09% WT), preventing the catastrophic background collapse observed in standard 3D Residual UNet (7.03% mean, 13.63% WT).
4. **Inter-Channel Projection Entanglement and Modality Fragility**: In standard 3D Vision Transformers, all four MRI sequences are jointly projected into patch tokens via an early convolutional layer. When individual sequences are omitted, the token representations suffer severe LayerNorm distribution shifts that cascade through multi-head self-attention, causing catastrophic collapse (4–11% mean Dice). Supervised fine-tuning with Random Modality Dropout ($p_{\text{drop}} = 0.25$) fails to prevent this collapse due to the scarcity of single-modality training steps and the linear entanglement of early weights. Architectural solutions requiring independent per-channel 3D patch embeddings aggregated via late attention pooling or gating are necessary to confer genuine channel-invariance. In contrast, 3D CNNs employ localized filters with additive channel accumulation and Instance Normalization, preserving substantial single-channel spatial activations (83.47% WT on FLAIR-only for nnU-Net).

6.2 Practical Model Selection Guidance

Based on our empirical results, we outline concrete practical considerations for model selection in multi-parametric 3D medical imaging:

- **Consider 3D VisReg JEPA when**:

1. *Annotations are extremely scarce*: In ultra-low label regimes (1% annotated volumes, corresponding to $\approx$ 11 patient scans), pre-training 3D VisReg JEPA on available unannotated cohorts and fine-tuning with a UNETR hybrid decoder provides the strongest performance (32.09% mean Dice, 50.09% WT), avoiding the optimization collapse of unregularized CNNs (7.03%).
2. *A hybrid decoding pathway is utilized*: When deploying ViT backbones, practitioners should prioritize hybrid convolutional decoders with multi-scale skip routing over pure FPN heads to ensure sharp boundary localization on small sub-compartments such as Enhancing Tumor.

- **Prefer Supervised Convolutional Baselines when**:
  1. *Moderate to full annotations exist ($\geq$ 5% labels)*: With at least 57 annotated patient volumes, deeply supervised 3D nnU-Net establishes superior accuracy across all anatomical compartments (88.19% WT, 83.00% TC, 67.98% ET) and significantly sharper boundary delineation (5.01–4.22 voxels HD95 on WT/TC).
  2. *Emergency triage with missing sequences is anticipated*: Convolutional architectures with localized feature channels provide vastly superior resilience when contrast-enhanced T1c (69.99% TC for nnU-Net vs. 8.00% for VisReg) or FLAIR sequences (83.47% WT for nnU-Net vs. 35.03% for VisReg) are unavailable.
  3. *Inference latency is critical*: Standard 3D Residual UNet is $3.7\times$ faster than UNETR Hybrid (260.13 ms vs. 986.92 ms per 128^3 volume).

6.3 Limitations and Methodological Validity

We explicitly delineate seven methodological limitations of this study:

1. **Generative Pre-Training Comparisons**: While this study includes strictly matched from-scratch Vision Transformer controls to isolate architectural vs. pre-training effects, we do not evaluate a 3D Masked Autoencoder (3D MAE) baseline under compute-matched schedules to directly contrast latent prediction against voxel-level reconstruction. Similarly, we evaluate VisReg as a representative single-encoder JEPA without an isolated fine-tuning comparison against SIGReg.
2. **Nature of Low-Data Background Collapse**: Standard 3D Residual UNet collapses to near-background predictions at 1% labels (7.03% mean Dice) under the matched 15-epoch budget (≈ 82 optimizer steps on 11 volumes at batch size 2). This appears to reflect an optimization and undertraining failure mode under severe class imbalance and short schedules rather than an intrinsic ceiling of convolutional architectures, as deeply supervised DynUNet achieves 28.05% mean Dice under the identical budget.
3. **Spatial Resolution and Preprocessing Coordinates**: Volumes are resampled from native $240 \times 240 \times 155$ ($1.0\,\text{mm}^3$) to a canonical $128 \times 128 \times 128$ grid following brain bounding box extraction. This introduces case-dependent spatial scaling (typically $\approx$ 1.2–1.5 mm per voxel across subjects), which is why reported HD95 surface distances are explicitly labeled in canonical grid voxels rather than physical millimeters to avoid conflating standardized array spacing with patient anatomy.
4. **Statistical Scope and Seed Variance**: All benchmark metrics reflect a single deterministic training run (seed $= 42$). The reported standard deviations ($\pm$) capture inter-subject anatomical variance across the $N = 242$ held-out test volumes, not variance across random initializations. Multi-seed evaluations with paired bootstrap confidence intervals are needed for formal statistical hypothesis testing.
5. **Baseline Training Budgets**: The 3D nnU-Net baseline implements the MONAI DynUNet architecture trained for a matched budget of 30 epochs. Full standalone nnU-Net pipelines train for 1000 epochs with extensive automated data augmentation and ensembling, representing a substantially higher compute regime.
6. **Uncalibrated Regularizer Weights**: VisReg weights are defaults ($\lambda_c = \lambda_{sc} = \lambda_{sh} = 1.0$) with no tuning sweep. The total pre-training loss couples the JEPA prediction term with three regularizers on different numerical scales, so the default weighting may over- or under-penalize collapse relative to anatomical signal. We deliberately do not rank methods

by Effective Rank: rank is reported only as a binary non-collapse check, and all comparative conclusions rest on downstream segmentation (Dice/HD95), where competitive accuracy at default weighting shows the encoder retained usable features.

7. **Early Joint Patch Projection Bottleneck**: The 3D ViT tokenization employs a single joint linear projection across all four multi-modal sequences. While standard in UNETR designs, this topology entangles input channels and induces acute sensitivity to omitted modalities during triage. Future architectures should evaluate decoupled, per-channel patch projections paired with permutation-invariant late attention pooling or gating.

6.4 Conclusion

In this paper, we investigated whether 3D Joint-Embedding Predictive Architectures (JEPA)—designed explicitly to avoid voxel-level reconstruction—can perform dense multi-compartment volumetric medical image segmentation. Through a systematic empirical evaluation on the BraTS 2024 GLI benchmark under the official three-region protocol (WT, TC, ET) comparing 3D VisReg JEPA against matched from-scratch ViT controls and supervised convolutional baselines (Residual UNet and deeply supervised DynUNet), we find that 3D JEPA successfully learns representations capable of dense multi-region segmentation, achieving $84.74 \pm 10.85\%$ WT, $75.77 \pm 19.92\%$ TC, and $58.83 \pm 32.34\%$ ET Dice (mean 73.11%) when paired with a UNETR hybrid decoder.

However, our empirical benchmark establishes critical architectural lessons and boundary conditions:

1. Fully supervised CNNs remain superior across all compartments (88.19% WT, 83.00% TC, 67.98% ET for nnU-Net), demonstrating that localized 3D convolutional priors outperform self-attention for volumetric dense prediction on full datasets.
2. Hybrid convolutional skip connections are essential to bridge the spatial granularity bottleneck of 16^3 patch tokenization, providing a $+5.44$pp mean Dice boost over pure ViT-FPN.
3. Under full supervision, self-supervised pre-training offers virtually zero advantage over from-scratch initialization (< 0.3pp difference across all compartments); its primary benefit is localized under extreme label scarcity (1% labels), where it jumpstarts optimization and prevents background collapse (32.09% vs. 7.03%).
4. Vision Transformers exhibit acute representation fragility under missing pulse sequences (4–11% mean Dice), whereas convolutional networks preserve substantial spatial redundancy (83.47% WT on FLAIR-only).

We conclude that 3D JEPA provides a viable foundation for volumetric medical imaging under extreme label scarcity, but requires hybrid spatial routing and channel-invariant mechanisms to match the boundary precision and clinical robustness of specialized convolutional networks.

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